INSTRUCTIONAL DESIGN PORTFOLIO

From technical content to applied learning.

Designing an online learning experience that helps developing medical-device auditors interpret evidence, investigate discrepancies and make defensible decisions.

Technical Learning Design

Fictional demonstration case study

This artefact is for demonstration and is not commissioned client work.

01 / PROJECT OVERVIEW

Starting with the learning problem.

Technical compliance training often contains dense terminology, regulatory requirements and process documentation. Presenting this information as slides alone may not give learners sufficient opportunity to practise applying it in realistic situations.

AUDIENCE

Medical-device auditors in training

FORMAT

Self-paced online learning

LEARNING NEED

Apply evidence-based audit reasoning

PROTOTYPE SCOPE

Five-screen learning sequence

The design challenge

How might technical source material be transformed into an accessible online experience that encourages learners to move beyond recalling information and begin applying it?

> **Design decision**
>
> Use short explanatory content, a realistic fictional discrepancy, guided practice and a scenario-based knowledge check with explanatory feedback.

My approach

This demonstration illustrates a proposed instructional design approach. It is not a reproduction of confidential work or a representation of a commissioned client project.

02 / LEARNING DESIGN

Define what learners should be able to do.

The following objectives are proposed for this fictional module. They would normally be validated against the course requirements, approved technical source material and subject-matter expert input.

Learning objectives

1. Identify relevant sources of documented evidence during a document-control audit.
2. Compare available evidence against applicable audit criteria.
3. Select an appropriate next investigative action when a documentation discrepancy is identified.

Instructional sequence

01 · EXPLAIN

Establish relevance and introduce document control.

02 · EXPLORE

Identify evidence an auditor might examine.

03 · PRACTISE

Investigate a fictional discrepancy.

04 · ASSESS

Select a defensible next action and review feedback.

Assessment strategy

A scenario-based question assesses decision-making rather than simple recall. Plausible distractors represent common mistakes, including jumping to conclusions or taking action without documenting the evidence.

> **Feedback principle**
>
> Explain why an answer is appropriate, or what the learner should reconsider, so the assessment also supports learning.

03 / STORYBOARD

Turning the design into screens.

SCREEN 01 · INTRODUCTION

The importance of document control

Auditors need to establish whether personnel are using the correct, approved version of a work instruction.

Learner action: Identify the purpose of document control.

Rationale: Establish relevance before introducing detailed terminology.

SCREEN 02 · CONCEPT EXPLORATION

What evidence should an auditor examine?

- Approved document register
- Revision identifier
- Approval records
- Document available at the point of use

Learner action: Compare the evidence with applicable criteria.

Rationale: Organise information in a scannable checklist.

SCREEN 03 · GUIDED PRACTICE

Investigate the discrepancy

An operator is using revision 2 of a work instruction, while the document register identifies revision 3 as approved. The reason for the discrepancy has not yet been established.

Learner action: Identify what needs verification.

Rationale: Introduce uncertainty that requires evidence-based reasoning.

04 / ASSESSMENT DESIGN

Assessing judgement through scenarios.

SCREEN 04 · KNOWLEDGE CHECK

What would you do next?

During a fictional audit, an operator is using revision 2 of a work instruction, while the document register identifies revision 3 as approved. You have not established why the discrepancy occurred.

A. Declare the entire quality management system non-compliant immediately.

B. Verify the discrepancy, establish the applicable criteria and gather relevant evidence.

C. Replace the instruction without documenting the discrepancy.

> **Correct response: B**
>
> Verify the document versions, determine the applicable criteria and establish the facts before deciding whether an audit finding is justified.

Why these distractors?

Option A represents premature judgement. Option C represents action without sufficient investigation or documentation. Option B requires the learner to prioritise evidence and relevant criteria before reaching a conclusion.

Accessibility considerations

- Use clear headings and concise instructions.
- Make interactive controls keyboard accessible.
- Provide readable colour contrast and visible focus states.
- Do not rely on colour alone to communicate correctness.
- Provide feedback in text rather than through visual cues alone.

05 / EVALUATION AND REFLECTION

How would I evaluate the design?

The module would be reviewed before release and evaluated after learners begin using it. The following is a proposed evaluation plan, not a report of measured outcomes.

Before launch

- Validate technical accuracy with a qualified subject-matter expert.
- Check alignment between objectives, content and assessment.
- Test navigation, keyboard access and mobile layouts.
- Review question wording and feedback for ambiguity.

After launch

- Review assessment performance against defined success criteria.
- Identify questions with unexpected response patterns.
- Gather learner feedback on clarity and relevance.
- Use findings to prioritise content and interaction improvements.

Reflection

The key design decision is to connect technical concepts to realistic learner decisions. Scenario-based practice provides a bridge between information presented in a module and the reasoning learners may need to apply in practice.

> **Scope and limitations**
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> This fictional artefact demonstrates design reasoning only. It does not establish regulatory compliance, demonstrate measured learning gains or replace approved technical guidance.